

IT NS 5 – INTRODUCTION TO NETWORK CAREER CERTIFICATION

Chapter I

Introduction to Networking

Introduction

Generally, networking/computer networking connects two or more computing devices that transport and exchange data in an information system. It also manages and maintains network infrastructure and policies. Types of networks are Local Area Networks, Metropolitan Area Networks, Wide Area Networks, and Personal Area Networks in the geographical areas that serve and have a core purpose. In addition, the OSI model has seven layers and the TCP/IP model has four layers that the computer systems use to communicate over a network. Networking plays a crucial role in different sectors, it enables everything that we do today. Without networking, business would not work, and our personal social life would feel like it stopped.

Learning Outcome

At the end of the unit, the student should be able to:

- *Identify networking benefits;*
- *Give example uses of networking;*
- *Differentiate between network types; and*
- *Compare and contrast TCP/IP vs OSI model.*

Learning Content

- Definition and importance of networking
- Types of networks (LAN, WAN, MAN, PAN)
- Networking Models (OSI Model, TCP/IP Model)

Teaching and Learning Activities

Introduction to Networking

A **networking**, also known as **computer networking**, is the practice of transporting and exchanging data between nodes over a shared medium in an information system. Networking comprises not only the design, construction and use of a network, but also the management, maintenance and operation of the network infrastructure, software and policies.



Networking is the process of connecting two or more computing devices, such as desktop computers, mobile devices, routers or application to enable the transmission and exchange of information resources.

Computer networking enables devices and endpoints to be connected to each other on a local area network LAN or to a larger network, such as the internet or a private wide area network WAN. This is an essential function for service providers, businesses and consumers worldwide to share resources, use or offer services, and communicate. Networking facilitates everything from telephone calls and text messaging to streaming video and the internet of things IoT.

Importance of Computer Networking

In the world of Information Technology (IT) information is building block for effective communication. Communication is medium that helps us to drive our day to day professional and personal operations. Where computer networking acts as base of everything as the best and most important IT solutions. Below is list of points that we do with the help of computer networks, or things that we get benefited with the help of computer networks or it has become possible or effective due to computer network.

- Provides best way of business communication.
- Streamline communication.
- Cost-effective resource sharing.
- Increase in efficiency.
- Allows File sharing.
- User communication.
- Cut back on costs.
- Cut costs in software.
- Cut costs in hardware.
- Overcoming geographic separation.

Computer networking has also played a crucial role in the development of new technologies like cloud computing, big data analytics, and the Internet of Things have all benefited greatly from computer networking (IoT). Businesses are now able to gather and analyze enormous volumes of data, automate their procedures, and make better choices faster thanks to these technologies.

In general, we can see “**Communication**” and “**Sharing**”, that is all networking is about in simple terms. It enables everything that we do today. In the current scenario, the business would not work without computer networking and our personal and social life will also feel like it has been stopped.

Uses of Computer Network

There are many different uses and applications for computer networks, including:

- **Resource sharing:** One of the primary uses of computer networks is to share resources, such as printers, scanners, and data storage devices. This allows multiple users to access the same resources, which can help to reduce costs and increase efficiency.
- **Communication:** Computer networks also facilitate communication between individuals and groups, whether through email, instant messaging, or video conferencing. This can be especially valuable for remote workers, or for teams that are spread out across different locations.
- **Collaboration:** Networks can enable collaboration between multiple users, allowing them to work together on the same project or document in real-time. This can be especially useful for teams that are working on complex projects or that need to coordinate their efforts closely.
- **Internet access:** Computer networks provide access to the internet, which is essential for many businesses and individuals. This allows users to browse websites, access online resources, and communicate with others around the world.
- **Data sharing:** Networks allow for the sharing of data and information, whether between users in the same organization or between different organizations. This can be valuable for data analysis, research, and other applications that require access to large amounts of information.
- **Security:** Computer networks can be used to implement security measures, such as firewalls and encryption, which can help to protect sensitive information and prevent unauthorized access.

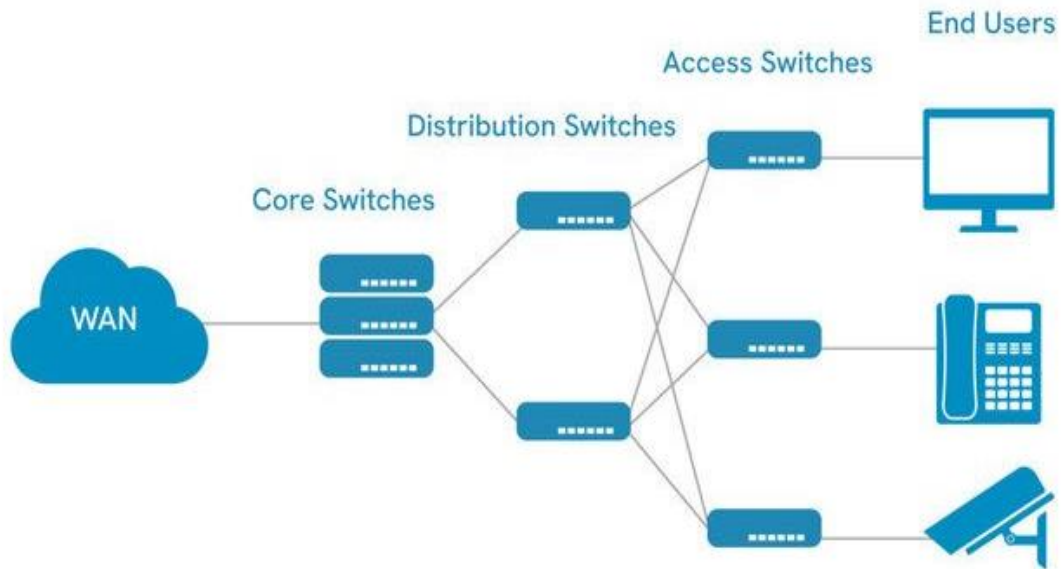
Types of Networks

There are several types of computer networks. The critical difference lies in the geographical areas they serve and their core purpose. These include:



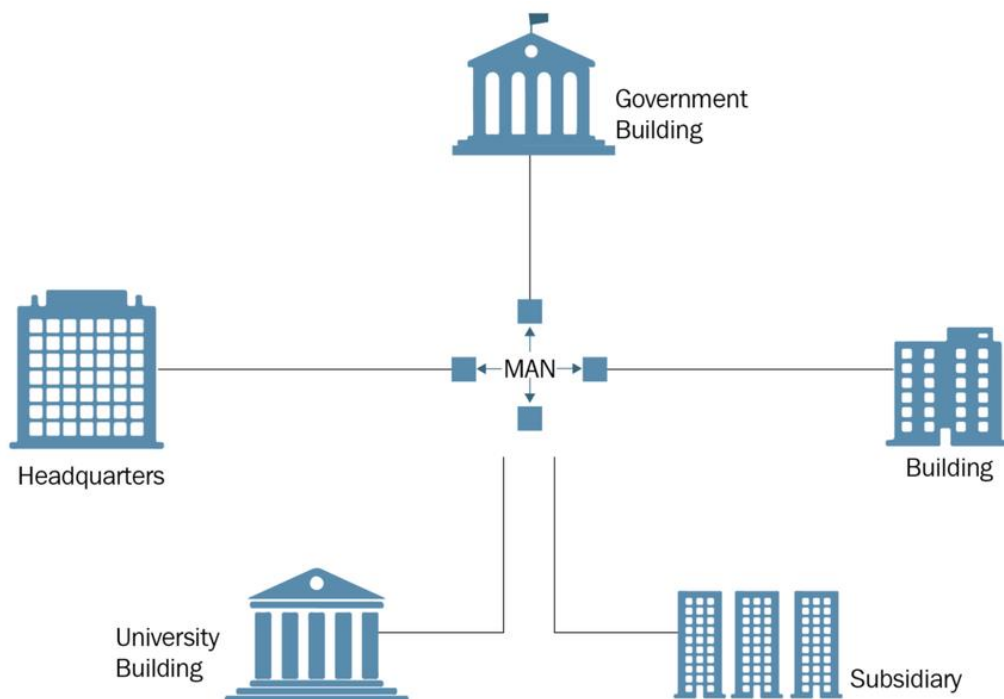
❖ Wide Area Network (WAN)

As the name implies, a WAN connects computers across large geographical areas, such as regions and continents. WANs often have collective or distributed ownership models for network management purposes. Cloud networks serve as one example, since they're hosted and delivered by public and private cloud infrastructure across the globe.



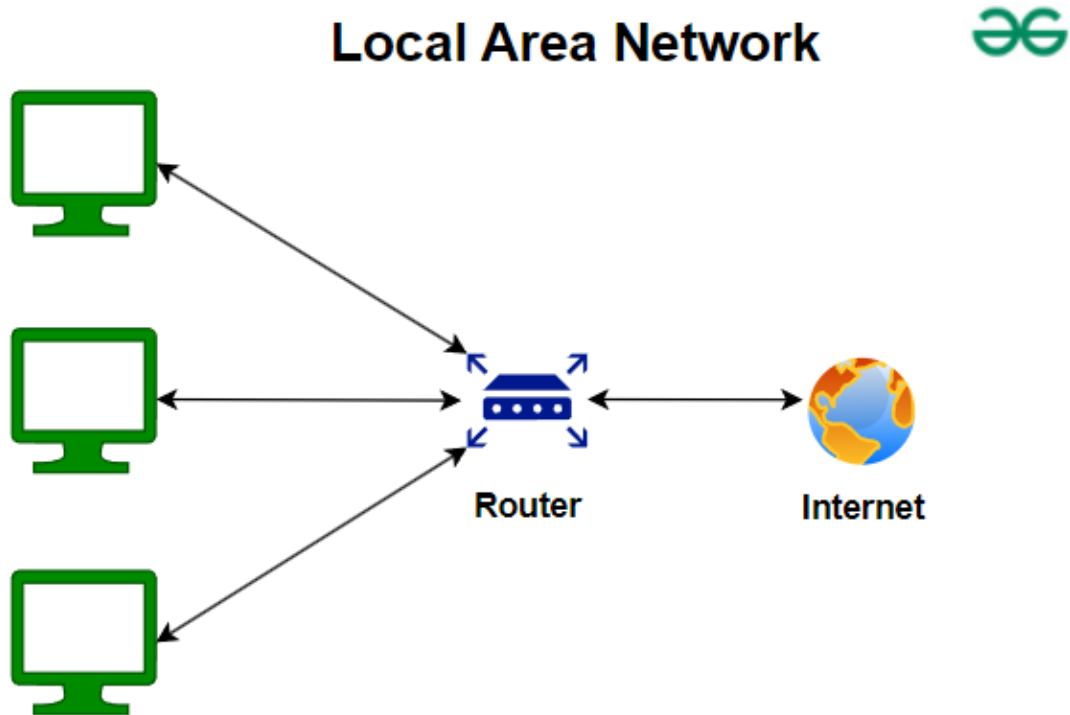
❖ Metropolitan Area Network (MAN)

MANs are larger than LANs but smaller than WANs. Cities and government entities typically own and manage MANs.



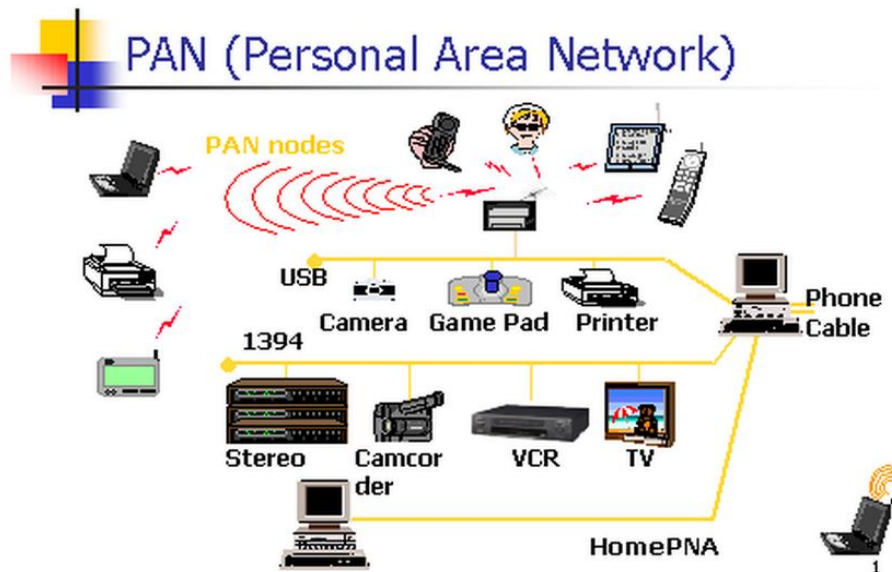
❖ Local Area Network (LAN)

A LAN connects computers over a relatively short distance, such as those within an office building, school or hospital. LANs are typically privately owned and managed.



❖ Personal Area Network (PAN)

A PAN serves one person. If a user has multiple devices from the same manufacturer (an iPhone and a MacBook, for instance), it's likely they've set up a PAN that shares and syncs content-text messages, emails, photos and more-across devices.



Networking Models

The **Open Systems Interconnection (OSI)** model describes seven layers that computer systems use to communicate over a network. It was the first standard model for network communications, adopted by all major computer and telecommunication companies in the early 1980s.

The modern Internet is not based on OSI, but on the simpler TCP/IP model. However, the OSI 7-layer model is still widely used, as it helps visualize and communicate how networks operate, and helps isolate and troubleshoot networking problems.

OSI was introduced in **1983** by representatives of the major computer and telecom companies, and was adopted by ISO as an international standard in 1984.

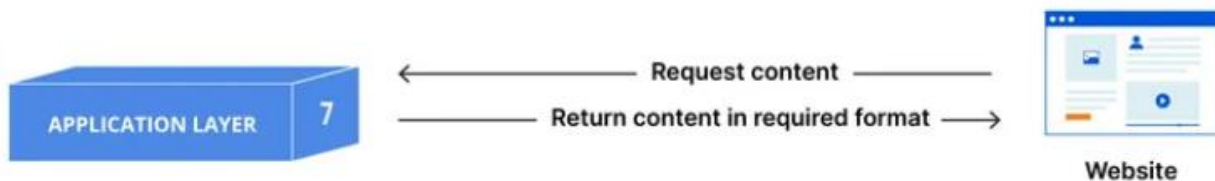
OSI MODEL: THE OSI 7 LAYERS

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

We'll describe OSI layers "**top down**" from the application layer that directly serves the end user, down to the physical layer.

Application Layer

The application layer is used by end-user software such as web browsers and email clients. It provides protocols that allow software to send and receive information and present meaningful data to users. A few examples of application layer protocols are the **Hypertext Transfer Protocol (HTTP)**, **File Transfer Protocol (FTP)**, **Post Office Protocol (POP)**, **Simple Mail Transfer Protocol (SMTP)**, and **Domain Name System (DNS)**.



Presentation Layer

The presentation layer prepares data for the application layer. It defines how two devices should encode, encrypt, and compress data so it is received correctly on the other end. The presentation layer takes any data transmitted by the application layer and prepares it for transmission over the session layer.



Session Layer

Session layer creates communication channels, called **sessions**, between devices. It is responsible for opening sessions, ensuring they remain open and functional while data is being transferred, and closing them when communication ends. The session layer can also set checkpoints during a data transfer—if the session is interrupted, devices can resume data transfer from the last checkpoint.



Transport Layer

The transport layer takes data transferred in the session layer and breaks it into “**segments**” on the transmitting end. It is responsible for reassembling the segments on the receiving end, turning it back into data that can be used by the session layer. The transport layer carries out flow control, sending data at a rate that matches the connection speed of the receiving device, and error control, checking if data was received incorrectly and if not, requesting it again.



Network Layer

The network layer has **two main functions**. One is breaking up segments into network packets, and reassembling the packets on the receiving end. The other is routing packets by discovering the best path across a physical network. The network layer uses network addresses (typically **Internet Protocol addresses**) to route packets to a destination node.



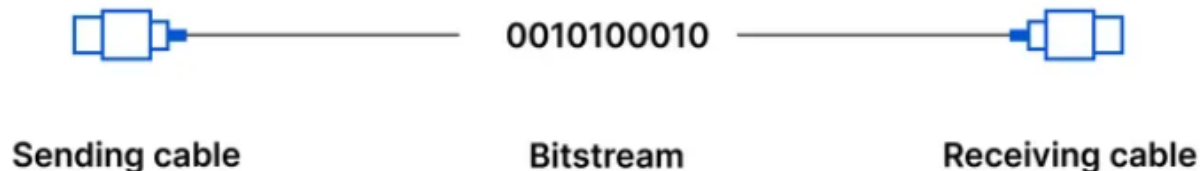
Data Link Layer

The data link layer establishes and terminates a connection between two physically-connected nodes on a network. It breaks up packets into frames and sends them from source to destination. **This layer is composed of two parts—Logical Link Control (LLC)**, which identifies network protocols, performs error checking and synchronizes frames, and **Media Access Control (MAC)** which uses MAC addresses to connect devices and define permissions to transmit and receive data.



Physical Layer

The physical layer is responsible for the physical cable or wireless connection between network nodes. It defines the connector, the electrical cable or wireless technology connecting the devices, and is responsible for transmission of the raw data, which is simply a series of 0s and 1s, while taking care of bit rate control.



Advantages of OSI Model

The OSI model helps users and operators of computer networks:

- Determine the required hardware and software to build their network.
- Understand and communicate the process followed by components communicating across a network.
- Perform troubleshooting, by identifying which network layer is causing an issue and focusing efforts on that layer.

TCP/IP MODELS

The **TCP/IP** model it stands for (**Transmission Control Protocol/Internet Protocol**) is a fundamental framework for computer networking, defining data transmission and ensuring reliable communication between devices. It consists of **four layers: The Link Layer, the Internet Layer, the Transport Layer, and the Application Layer**. Each layer has specific functions that help manage different aspects of network communication, making it essential for understanding and working with modern networks.

TCP/IP was designed and developed by the **Department of Defense (DoD)** in the **1960s**, it is a concise version of the OSI model and was developed alongside **ARPANET**, which became the foundation of the modern internet.

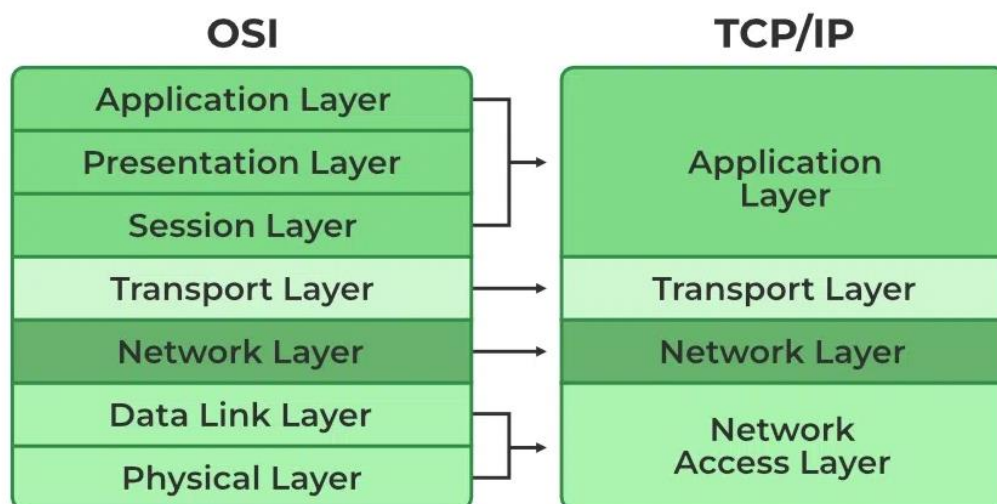
What does TCP/IP do?

The main work of **TCP/IP** is to transfer the data of computer from one device to another ensuring reliability and accuracy. It divides data into packets and combines them at the other end, maintaining data accuracy. This model is used in real-world internet environments, allowing flexibility in adapting to different physical implementations, rather than specifying a specific Physical Layer.

Difference between TCP and IP

Feature	TCP (Transmission Control Protocol)	IP (Internet Protocol)
Purpose	Ensures reliable, ordered, and error-checked delivery of data between applications.	Provides addressing and routing of packets across networks.
Type	Connection-oriented	Connectionless
Function	Manages data transmission between devices, ensuring data integrity and order.	Routes packets of data from the source to the destination based on IP addresses.
Error Handling	Yes, includes error checking and recovery mechanisms.	No, IP itself does not handle errors; relies on upper-layer protocols like TCP.
Flow Control	Yes, includes flow control mechanisms.	No
Congestion Control	Yes, manages network congestion.	No
Data Segmentation	Breaks data into smaller packets and reassembles them at the destination.	Breaks data into packets but does not handle reassembly.
Header Size	Larger, 20-60 bytes	Smaller, typically 20 bytes
Reliability	Provides reliable data transfer	Does not guarantee delivery, reliability, or order.
Transmission Acknowledgment	Yes, acknowledges receipt of data packets.	No

The diagrammatic comparison of the TCP/IP and OSI model is as follows:



Application Layer

The application layer is responsible for communication between sender and receiver, ensuring data reaches the receiver error-free and protecting upper-layer applications from data complexity.

It includes protocols like **HTTP and HTTPS**, which manage communication between web browsers and servers. **SSH, or Secure Shell**, is a terminal emulation software that maintains encrypted connections over TCP/IP sessions, ensuring data security and privacy.

The final application layer protocol is **NTP**, which synchronizes clocks on devices with a standard time source, making it useful in banking transactions.

Transport Layer

Data is split into packets, each of specific size, and must arrive at their destination and reassemble to form the full message. The transport layer exchanges data receipt acknowledgments and retransmits missing packets to ensure order without being corrupted. **Two protocols are used at the transport layer:**

TCP and UDP. TCP transmits data like character-by-character, allowing devices to communicate as if they were physically connected in a circuit.

UDP stands for User Datagram Protocol provides delivery services for datagrams, reducing connection validation for smaller data packets because it eliminates the process of validating a connection between the devices.

Internet or Network Layer

The network layer defines protocols that head the logical data transmission over a network. The first is IP, or Internet Protocol, which looks at the headers of each data packet and reads the IP address listed on it. That is the packet's destination, and the IP protocol then delivers the packet to the device with that address.

The second protocol is ICMP, Internet Control Message Protocol, which provides people with information about network problems they are experiencing. The final protocol is ARP, Address Resolution Protocol, which finds the device's hardware address using its known IP address.

Network Access Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver.

The packet's network protocol type, in this case, TCP/IP, is identified by network access layer. Error prevention and "framing" are also provided by this layer. **Point to Point Protocol (PPP)** framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

Other Common Internet Protocols

TCP/IP Model covers many Internet Protocols. The main rule of these Internet Protocols is how the data is validated and sent over the Internet. Some Common Internet Protocols include:

- **HTTP (Hypertext Transfer Protocol):** HTTP takes care of Web Browsers and Websites.
- **FTP (File Transfer Protocol):** FTP takes care of how the file is to be sent over the Internet.
- **SMTP (Simple Mail Transfer Protocol):** SMTP is used to send and receive data.

Difference between TCP/IP and OSI Model

TCP/IP	OSI
TCP refers to Transmission Control Protocol.	OSI refers to Open Systems Interconnection.
TCP/IP uses both the session and presentation layer in the application layer itself.	OSI uses different session and presentation layers.
TCP/IP follows connectionless a horizontal approach.	OSI follows a vertical approach.
The Transport layer in TCP/IP does not provide assurance delivery of packets.	In the OSI model, the transport layer provides assurance delivery of packets.
Protocols cannot be replaced easily in TCP/IP model.	While in the OSI model, Protocols are better covered and are easy to replace with the technology change.
TCP/IP model network layer only provides connectionless (IP) services. The transport layer (TCP) provides connections.	Connectionless and connection-oriented services are provided by the network layer in the OSI model.

Advantages of TCP/IP Model

- **Interoperability:** The TCP/IP model allows different types of computers and networks to communicate with each other, promoting compatibility and cooperation among diverse systems.
- **Scalability:** TCP/IP is highly scalable, making it suitable for both small and large networks, from local area networks (LANs) to wide area networks (WANs) like the internet.
- **Standardization:** It is based on open standards and protocols, ensuring that different devices and software can work together without compatibility issues.
- **Flexibility:** The model supports various routing protocols, data types, and communication methods, making it adaptable to different networking needs.
- **Reliability:** TCP/IP includes error-checking and retransmission features that ensure reliable data transfer, even over long distances and through various network conditions.

Disadvantages of TCP/IP Model

- **Complex Configuration:** Setting up and managing a TCP/IP network can be complex, especially for large networks with many devices. This complexity can lead to configuration errors.
- **Security Concerns:** TCP/IP was not originally designed with security in mind. While there are now many security protocols available (such as SSL/TLS), they have been added on top of the basic TCP/IP model, which can lead to vulnerabilities.
- **Inefficiency for Small Networks:** For very small networks, the overhead and complexity of the TCP/IP model may be unnecessary and inefficient compared to simpler networking protocols.
- **Limited by Address Space:** Although IPv6 addresses this issue, the older IPv4 system has a limited address space, which can lead to issues with address exhaustion in larger networks.
- **Data Overhead:** TCP, the transport protocol, includes a significant amount of overhead to ensure reliable transmission. This can reduce efficiency, especially for small data packets or in networks where speed is crucial.

TCP/IP model is the backbone of modern internet communication, allowing different devices and networks to connect and share information reliably. Despite some complexity and security concerns, its flexibility, scalability, and widespread adoption make it essential for both small and large networks. Overall, the TCP/IP model is crucial for ensuring efficient and effective network communication.

Assessment Task

After finishing the Learning Content, students should be able to answer the following exercises and activities.

Question 1: What is networking?

Question 2: Give the types of network?

Question 3: Why networking is crucial in different sectors?

Question 4: How many layers are there in OSI Model?

Question 5: How many layers are there in TCP/IP model?

Now that you have answered confidently the questions above, you are now ready to take the quiz. Login to your Canvas Student app in your computer or phone and take the quiz, or contact me if you have problem taking the online quiz.

Recommended learning materials and resources for supplementary reading.

- ✓ <https://www.youtube.com/watch?v=WexBQ1XgaDw>
- ✓ https://www.youtube.com/watch?v=4_zSIXb7tLQ
- ✓ <https://www.youtube.com/watch?v=ILErlaRp5cs>

Flexible Teaching Learning Modality (FTLM) adopted

Online (synchronous)

- Zoom Meeting as scheduled
- Messenger Application

Remote (asynchronous)

- Canvas Student Application. Instructions can be viewed at www.erickabuzo.com/class
- Printed Module

References

IBM (2021) *What is Computer Networking?* <https://www.ibm.com/topics/networking>

Kirvan, P. and Scarpatti, J. (2024) *What Is computer Networking?: Definition from TechTarget, Networking.* <https://www.techtarget.com/searchnetworking/definition/networking>

Learning Center (2023) *What is OSI Model: 7 Layers Explained: Imperva* <https://www.imperva.com/learn/application-security/osi-model/>